

AMENDMENT NO. 1 JANUARY 2017
TO
IS 5182 (PART 6) : 2006 METHOD FOR MEASUREMENT
OF AIR POLLUTION

PART 6 OXIDES OF NITROGEN

(First Revision)

[Page 4, clause 11.3, (V_a) and (V_t)] — Substitute the following for the existing

' V_a = volume of air sample, m³;

V_t = aliquot taken for analysis, ml'

(Reaffirmed 2022)

(Reaffirmed - 2012)

(Reaffirmed 2017)

भारतीय मानक

वायु प्रदूषण मापने की पद्धति

भाग 6 नाइट्रोजन के ऑक्साइड

(पहला पुनरीक्षण)

Indian Standard

METHOD FOR MEASUREMENT OF AIR POLLUTION

PART 6 OXIDES OF NITROGEN

(First Revision)

ICS 13.040.20

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

November 2006

Price Group 3

Environment Protection and Waste Management Sectional Committee, CHD 32

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Environment Protection and Waste Management Sectional Committee had been approved by the Chemical Division Council.

The standard was originally published in 1975. Based on the technological development in this area since last three decades, the committee decided to revise the standard. In this revision, details of the sampling apparatus and required glassware; precision and accuracy and also 4 h sampling have been incorporated.

The method is generally specific for nitrogen dioxide. In high temperature combustion processes, some other oxides of nitrogen and yielded which further oxidize to nitrogen dioxide in air at ordinary temperatures. So, the method is employed to determine the nitrogen oxide contents in the atmosphere as nitrogen dioxides.

There is no ISO Standard on this subject. In the preparation of this standard, considerable assistance has been derived from the following publications.

- a) CHRISTIE (A A) *et. al.* Field method for the determination of nitrogen dioxide in air. Analyst, 95 : 510-524 (1970).
- b) MAGESON, (J H) *et. al.* Evaluation of the sodium arsenite method for measurement of NO₂ in ambient air. J.A.P.C.A., 27 : 553-556 (1977).
- c) JACOB, (M B) and HOCHHEISER (S). Continuous sampling and untra-micro determination of nitrogen dioxide in air. Anal. Chem. 30 : 426 (1958).

The composition of the Committee responsible for formulating this standard is given in Annex A.

In reporting the results of a test or analysis made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'.

Indian Standard
METHOD FOR MEASUREMENT OF AIR POLLUTION
PART 6 OXIDES OF NITROGEN
(First Revision)

1 SCOPE

1.1 This standard prescribes a method for measurement of nitrogen oxides (as NO_2) in ambient air.

1.2 This method is applicable for 4 to 8 h integrated sampling of NO_2 in ambient air.

2 REFERENCES

The following standards contain provisions which through reference in this text constitute provisions of this standard. At the time of publication the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

<i>IS No.</i>	<i>Title</i>
1070 : 1992	Reagent grade water (<i>third revision</i>)
2303 (Part 1/ Sec 1) : 1994	Grading glass for alkalinity: Part 1 Hydrolytic resistance, Section 1 Hydrolytic resistance of glass grains at 98°C — Method of test and classification (<i>first revision</i>)
4167 : 1980	Glossary of terms relating to air pollution (<i>first revision</i>)

3 TERMINOLOGY

The definitions given in IS 4167 shall apply.

4 PRINCIPLE

Ambient nitrogen dioxides are collected by bubbling air through a solution of sodium hydroxide and sodium arsenite. The concentration of nitrite ion (NO_2^-) produced during sampling is determined colorimetrically by reacting the nitrite ion with phosphoric acid, sulphanilamide and N-(1-naphthyl)-ethylenediamine di-hydrochloride (NEDA) and measuring the absorbance of the highly coloured azo-dye at 540 nm.

5 RANGE

The nominal range of the method is 6 to 750 $\mu\text{g NO}_2/\text{m}^3$

(0.003 to 0.4 ppm). The range of the analysis is 0.04 to 2.0 $\mu\text{g NO}_2/\text{ml}$ following Beer's Law throughout this range (0 to 1.0 absorbance units). Under the specified conditions of 50 ml of absorbing reagent, a sampling rate of 200 cm^3/min for 24 h and a sampling efficiency of 82 percent, the range of the method is, therefore, 6 to 420 $\mu\text{g NO}_2/\text{m}^3$ (0.003 to 0.22 ppm). Nitrogen dioxide concentrations in the range of 420 to 750 $\mu\text{g}/\text{m}^3$ (0.22 to 0.4 ppm) are accurately measured by 1 : 1 dilution of the collected sample.

6 INTERFERENCES

6.1 Nitric oxide (NO) is a positive interferant and carbon dioxide (CO_2) is a negative interferant. The average error resulting from normal ambient concentrations on NO and CO_2 is small for most monitoring situations and does not necessitate applying a correction to measurements obtained with the method.

6.2 Potential interference from sulphur dioxide (SO_2) is eliminated by converting any SO_2 to sulphate using hydrogen peroxide during analysis.

7 SAMPLE PRESERVATION

Collected samples are stable for at least six weeks at room temperature $25 \pm 2^\circ\text{C}$. Stored samples should be tightly sealed to prevent absorption of NO_2 from the atmosphere.

8 APPARATUS

8.1 **Sampler** — A sampler consists of the following components.

8.1.1 *Sample Inlet* — Teflon or glass tube with an inverted teflon or glass funnel at the sampling point to prevent entrance of precipitation.

8.1.2 *Absorber* — Standard glass midjet impinger shall be used for sampling purpose. The glass impingers shall be made from Class HGB 3 glass conforming to IS 2303 (Part 1/Sec 1). Dimensional details of midjet impinger shall be as given in Fig. 1.

8.1.3 *Moisture Trap* — The outlet impinger, a moisture trap made up of glass or polypropylene tube loosely packed

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9.7 Phosphoric Acid, 85 percent.

9.8 Sodium Arsenite

CAUTION — Arsenic compounds are highly toxic and should be handled with extreme care. Avoid contact with skin and especially with eyes. Avoid generating dust or breathing dust. Keep away from food. Wash hands after handling it. Do not take internally.

9.9 Sodium Hydroxide

9.10 Sodium Nitrite — Assay of 97 percent NaNO_2 or greater.

9.11 Sodium Nitrite Stock Solution (1 000 $\mu\text{g NO}_2/\text{ml}$)

— Dissolve 1.500 g of desiccated sodium nitrite in distilled water and dilute to 1 000 ml such that a solution containing 1 000 $\mu\text{g NO}_2/\text{ml}$ is obtained. The amount of NaNO_2 to be used if the assay percent is less than 100 percent, is calculated as follows:

$$G = 1.500/A$$

where

G = amount of NaNO_2 in g;

1.500 = gravimetric conversion factor; and

A = assay, percent (should be 97 or greater).

NOTE — This stock solution can be stored for six weeks, if refrigerated.

9.11.1 *Sodium Nitrite Solution* (10 $\mu\text{g NO}_2/\text{ml}$) — Pipette 5 ml of the stock solution (see 9.11) in to a 500 ml volumetric flask and dilute to volume with distilled water.

9.11.2 *Sodium Nitrite Solution* (1 $\mu\text{g NO}_2/\text{ml}$) — Pipette 25 ml of the solution (see 9.11.1) in to a 250 ml volumetric flask and dilute to volume with absorbing reagent solution (see 9.2), prepare fresh daily.

9.12 Sulphanilamide — Melting point 165 to 167°C.

9.13 *Sulphanilamide Solution* — Dissolve 20.0 g of sulphanilamide in 700 ml of distilled water. Add with mixing 50 ml of 85 percent phosphoric acid and dilute to 1 000 ml.

NOTE — This solution is stable for one month, if refrigerated.

10 PROCEDURE

10.1 Sample Collection

10.1.1 *Four Hourly Sampling*

10.1.1.1 Assemble the sampling apparatus at the sampling site. Components upstream from the absorber may be connected where required with teflon/silicon tubing;

glass tubing with dry ball joints; or glass tubing with butt-to-butt joints with teflon or polypropylene. Add exactly 30 ml of absorbing reagent to the calibrated absorber. Disconnect the funnel. Connect the calibrated flow meter, measure the flowrate before sampling and record as F_i . Start sampling only after obtaining an initial flowrate of 1 litre per minute. Ensure packing of ice around bubbler for low temperature to improve absorption efficiency. For effective cooling thermoelectric cooling with thermostat to maintain temperature of absorbing reagent at 10-15°C should be used for better absorption efficiency.

10.1.1.2 *Sample for 4 hours*

Record the exact sampling time by recording initial time (t_i) and final time (t_f) of sampler. Measure and record the flowrate (f_i) before the sampling and flowrate (f_f) after the sampling. Seal the collected samples after making it up to 30 ml using distilled water and transport to the laboratory for analysis.

10.1.2 *24-Hourly Sampling*

10.1.2.1 Assemble the sampling apparatus at the sampling site. Components upstream from the absorber may be connected, where required, with teflon/silicon tubing; glass tubing with dry ball joints; or glass tubing with butt-to-butt joint with tygon, teflon or polypropylene.

Add exactly 30 ml of absorbing reagent to the calibrated absorber. Disconnect the funnel, connect the calibrated flow meter, measure the flowrate before sampling and record as F_i . Start sampling only after obtaining an initial flowrate in this range. Sampling rate need to be chosen depending upon expected range of concentration.

10.1.2.2 *Sample for 24 hours*

Record the exact sampling time in minutes by recording initial (t_i) and final (t_f) time of sampling. Measure and record the flowrate before (f_i) and after (f_f) the sampling. Seal the collected samples after making it up to 30 ml using distilled water and transport to the laboratory for analysis.

10.2 Analysis

10.2.1 Pipette 10 ml of the collected sample into 50 ml volumetric flask. Now add 1 ml of hydrogen peroxide solution, 10 ml of sulphanilamide solution, and 1.4 ml of NEDA solution using pipette, with thorough mixing after the addition of each reagent and make up to 50 ml with distilled water.

10.2.2 Prepare a blank in the same manner using 10 ml of un-exposed absorbing reagent. After a 10 min colour development interval, measure and record the absorbance

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at 540 nm against the prepared reagent blank. Determine NO_2 from the calibration curve (see 12.3). Samples with an absorbance greater than 1.0 must be reanalyzed after diluting an aliquot of the collected samples with an equal quantity of un-exposed absorbing reagent.

NOTES

- 1 A randomly selected 5-10 percent of the samples should be reanalyzed as apart of an internal quality assurance program.
- 2 A known concentration of NO_2 solution may analyze with sample as an internal quality assurance program.

11 CALCULATIONS

11.1 Air Volume

Calculate the volume of air drawn for sample as follows:

$$V_a = \frac{f_i + f_f}{2} \times (t_f - t_i) \times 60 \times 10^{-3}$$

where

- V_a = volume of air sample, m^3 ;
- f_i = air flowrate before sampling, litre per minute;
- f_f = air flowrate after sampling, litre per minute;
- t_i = initial time in hours;
- t_f = final time in hours;
- 10^{-3} = conversion of litre to m^3 ; and
- 60 = conversion of hours to min.

If the temperature and pressure conditions at the time of the initial and final air flowrate measurements are substantially different from the condition under which the flow meter was calibrated. appropriate corrections to the flowrate measurements may be made to improve the accuracy of the resultant NO_2 concentration measurement. The mathematical form of these correction depends on the type of flow meter used; consult an appropriate reference for guidance.

11.2 NO_2 Concentration in Analyzed Sample

Determine $\mu\text{g NO}_2/\text{ml}$ graphically from the calibration curve or compute from the slope and intercept values.

11.3 NO_2 Concentration in Air Sample

Calculate NO_2 as μg per cubic metre of air as follows:

$$\text{NO}_2 \mu\text{g}/\text{m}^3 = \frac{\text{NO}_2 \times D \times V_s}{V_a \times 0.82 \times V_t}$$

where

- NO_2 = NO_2 concentration in analyzed sample, μg ;
[Graph Factor $\times$ (Sample Abs.-Blank Abs)]
- D = dilution factor ($D = 1$ for no dilution : $D = 2$ for 1 : 1 dilution);
- V_s = final volume of sampling solution, ml;

- V_a = volume of air sample, ml;
- 0.82 = sampling efficiency; and
- V_t = aliquot taken for analysis.

11.4 The NO_2 concentration as ppm shall be calculated as follows:

$$\text{NO}_2, \text{ppm} = (\text{NO}_2 \mu\text{g}/\text{m}^3) \times 5.32 \times 10^{-4}$$

12 CALIBRATION

12.1 Flow Meter

Calibrate the flow meter against a calibrated standard flow measurement device such as a wet test meter, bubble flow meter, or other reliable volume measurement standard. Calibrate in units of standard cm^3/min , that is, correct to 25°C and 760 torr.

12.2 Absorber

Calibrate midjet glass impinger as follows:

- a) Pipette 35 ml distilled water absorbing reagent using 5 ml calibrated pipette and check correctness of marking on impinger.
- b) Use 3 mm thickness metal disk and check gap between bottom and tip of the impinger. 3 mm disk must not be loose after fitting head of impinger.
- c) Use three drill bits of 0.9 mm 1.0 mm and 1.1 mm and check bore size of nozzle. Bore size has to be 1.0 mm with tolerance of 0.1mm. Impingers must be cleaned thoroughly fortnightly and also before attempting calibration. Grease must be put on mouth of impinger while using.

12.3 Spectrophotometer

12.3.1 Prepare calibration curve using $1 \mu\text{g}/\text{ml}$ working standards (see 9.11.2) in accordance with the analytical procedure given in 12.3.2. Measure and record the absorbance for each calibration standard (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20 $\mu\text{g NO}_2$).

12.3.2 Plot absorbance (y-axis) versus the corresponding concentration in $\mu\text{g NO}_2/50 \text{ ml}$ solution (x-axis). Draw or compute the straight line best fitting the data to obtain the calibration curve.

13 PRECISION AND ACCURACY

Based on results from a collaborative study the within laboratory standard deviation is $8 \mu\text{g}/\text{m}^3$ (0.004 ppm) and the between laboratory standard deviation is $11 \mu\text{g}/\text{m}^3$ (0.006 ppm) over the range of 50 to $300 \mu\text{g NO}_2/\text{m}^3$ (0.027 to 1.16 ppm). Based on results from a collaborative study the method has an average bias of 3 percent over the range of 50 to $300 \mu\text{g NO}_2/\text{m}^3$ (0.027 to 0.16 ppm).

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ANNEX A (Foreword)

COMMITTEE COMPOSITION

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Amendments Issued Since Publication

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